

Total no. of Pages:

Roll no.

Degree: B.TECH Semester: 3 — Course work

MID-SEMESTER EXAMINATION, JUNE 2026

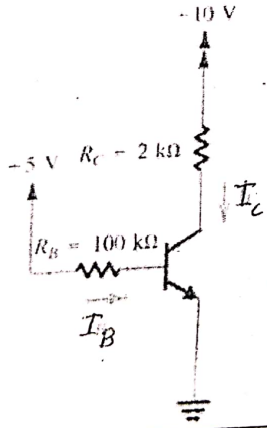
Course Title: Microelectronics circuits and Applications

Course Code: EAEP304

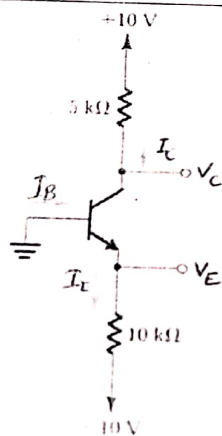
Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Draw and explain the Hybrid- π and T models of BJT.	2	CO1
1b	What is meant by transconductance (g_m) of a MOSFET?	2	CO1
2a	Explain the biasing techniques used in IC amplifiers.	2	CO1
2b	Analyse the high-frequency response of a Common Source (CS) amplifier with passive load.	2	CO1
3a	Define voltage gain, current gain and power gain.	2	CO1
3b	Explain the operation of a Cascode amplifier and mention its advantages.	2	CO2
4a	Analyse the BJT circuit shown below. Assume that the transistor has a current gain $\beta = 100$ and that the base-emitter junction voltage is $V_{BE} = 0.7\text{ V}$. Determine the voltages at all nodes and the currents in all branches. 	2	CO2
4b	In the circuit shown below, the voltage at the emitter was measured and found to be -0.7 V . If $\beta = 50$, find I_E , I_B , I_C and V_C .	2	CO2

Cont...-



5a

Compare BJT differential pair and MOS differential pair.

2

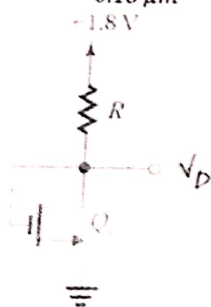
CO2

5b

For the circuit in figure given below, find the value of R that results in $V_D = 0.8V$. The MOSFET has $V_{in} = 0.5V$, $\mu_n C_{ox} = 0.4 \text{ mA/V}^2$, $W/L = \frac{0.72 \mu m}{0.18 \mu m}$ and $\lambda = 0$.

2

CO2



(where, $\mu_n C_{ox}$ is the process transconductance parameter of the NMOS transistor and λ is the channel-length modulation parameter).